

SLC Agricola

Softening Commodities - Assuming Coverage with N Rating and 2013 PT of R\$26

We assume coverage of SLC Agricola (SLCE3) with a Neutral rating and 2013 price target of R\$26 (under prior J.P. Morgan coverage SLCE3 was rated Neutral with a Dec-12 PT of R\$19). SLC is the best vehicle to play the soft-commodities and land businesses in Brazil. The company has state-of-the-art technology, allowing one of the highest yields globally and its accretive land business has room to appreciate alongside rising land prices. Our Neutral rating is based on a stretched valuation and a more cautious outlook on grain prices which limits margin expansion potential.

- **SLC Agricola should improve yields with investments in technology and expansion of planted area.** SLCE3 has historically been an industry leader with yields/ha for soy, corn and cotton above market averages. We believe the company will continue to outperform in this area given its constant investments in technology (R&D and Machinery), its plan to purchase and transform new land, and the geographic diversity of its farms. We assume acreage expands at a 5% CAGR (2012-20) and yields expand at 100 bps on a yearly basis which will allow the company to generate an adjusted EBITDA margin of close to 30%.
- **P/NAV of 0.83x, below average of ~1.1x since 2007, suggests value in an environment of rising land prices.** Furthermore, an increase in land value would directly benefit the company's land transformation business. The formation of SLC LandCo positions the company to continue to invest, unlock value from its existing land holdings and benefit from appreciating land in the long term.
- **Diversified crops mitigate weather volatility.** SLC's crop diversification helps maintain stable cash generation in case of poor weather conditions. Historically, SLC has been able to keep EBITDA margin close to 30%, despite the weather volatility inherent to the business which consequently causes swings in the price of commodities.
- **JPM is more bearish on soft-commodities prices.** After a brief recovery in prices through the end of 2012, our global commodities team is more cautious on prices of soy and corn, which we believe narrows the company's ability to expand margins. Considering a more cautious view for commodities, we believe SLC will be able to maintain current adjusted EBITDA margins. Our sensitivity analysis shows that any 10% increases in grains/cotton prices likely would bring EBITDA margin to 35%.
- **Valuation is stretched.** Our price target is derived from a blended valuation between (i) DCF from the farming business and (ii) NAV from its land and facilities. We arrive at a Dec-13 target of R\$26 per share, or 28% upside from current levels.

SLC Agricola S.A. (SLCE3.SA;SLCE3 BZ)

FYE Dec	2011A	2012E	2013E	2014E	2015E
EBITDA (R\$ mn)					
FY	464	211	295	346	381
Revenues FY (R\$ mn)	1,006	1,192	1,444	1,434	1,558
ROE FY	14.0%	1.6%	6.5%	8.7%	8.6%
EPS Reported FY (R\$)	3.01	0.36	1.55	1.85	2.13

Source: Company data, Bloomberg, J.P. Morgan estimates. 'Bloomberg' above denotes Bloomberg consensus estimates.

Neutral

SLCE3.SA, SLCE3 BZ

Price: R\$20.37

Price Target: R\$26.00

LatAm Capital Goods & Agribusiness

Cassio Lucin ^{AC}

(55-11) 4950-3893

cassio.lucin@jpmorgan.com

Thomas McElwee

(55-11) 4950-6719

thomas.mcelwee@jpmorgan.com

Banco J.P. Morgan S.A.

Price Performance

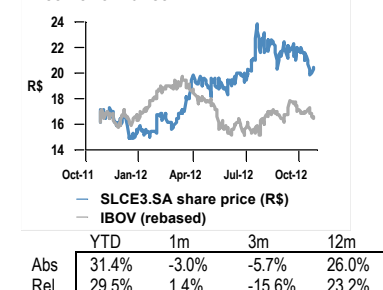


Table 1: JPMe Multiples

Ticker	P/BV 2012	P/BV 2013	EV/ EBITDA 2012	EV/ EBITDA 2013
SLCE3	1.04	0.95	11.7	8.0
AGRO US	1.02	0.95	9.3	6.2
CRESY US	1.00	0.90	n.a.	n.a.

Source: J.P. Morgan estimates. Price: 10/25/12 close.

Table 2: 2013 JPMe vs BBG Consensus

	JPMe	BBGe	%
Net Revenue	1,444	1,202	20%
EBITDA	295	282	5%
EV/EBITDA	8.0	10.1	16%

Source: Bloomberg, J.P. Morgan estimates.

Please see the most recent [Agricuture Weekly](#) from our **Commodities Analyst Pierre-Henri Dietz** for more information on JPMe commodities outlook.

Company Data

Price (R\$)	20.37
Date Of Price	25 Oct 12
52-week Range (R\$)	24.80 - 14.49
Mkt Cap (R\$ mn)	2,014.54
Fiscal Year End	Dec
Shares O/S (mn)	99
Price Target (R\$)	26.00
Price Target End Date	31 Dec 13

See page 25 for analyst certification and important disclosures, including non-US analyst disclosures.

J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

Table of Contents

Investment Thesis	3
Risks to Rating and Price Target	6
Company Description	6
Company Overview	7
Financial Outlook	10
Valuation	11
Appendix I: Price Outlook for Key Commodities.....	13
Appendix II: Overview of Grains and Cotton	14
Supply-Side Analysis Should Drive Investment Decisions	14
Grains – Driven by Planted Area and Crop Yields.....	14
Cotton – Off Historical Peak.....	21

Investment Thesis

SLC Agricola (SLCE3)

Neutral

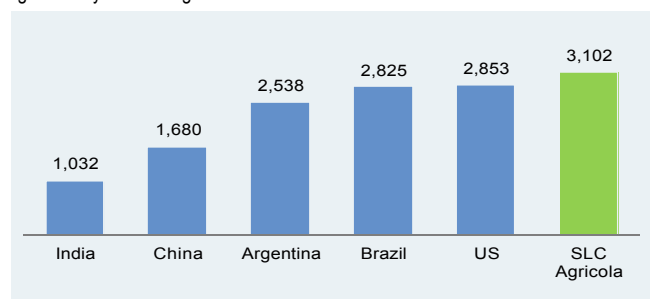
Investments in technology, expansion in planted area and geographic diversity should allow the company to improve yields.

Yields should continue to improve from investments in technology, expansion of planted area and the geographic diversity of farms

SLC has a proven track record of utilizing best agricultural practices to improve the agricultural yields of its farms (Figures 1-3). These yields, coupled with the company's large scale which dilutes fixed costs and gives the company bargaining power with suppliers and clients, have allowed the company to deliver an average EBITDA margin of ~30% for 2008-2011, adjusting for the value of biological assets. SLC has a Capex plan to (i) invest in technology and other improvements and (ii) acquire more land (planted area) that should allow the company to improve these yields moving forward. Finally, the company currently operates 14 diversified farms spread across the *cerrado brasileiro*, which allows the company to harvest and process grains for most of the year.

Figure 1: Global Soybean Yields vs. SLCE3

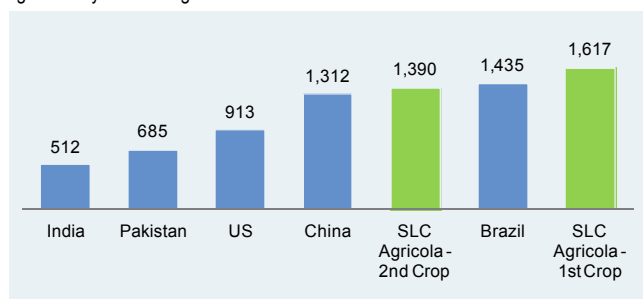
Kg/ha – 5-year average



Source: Company filings.

Figure 2: Global Cotton Yields vs. SLCE3

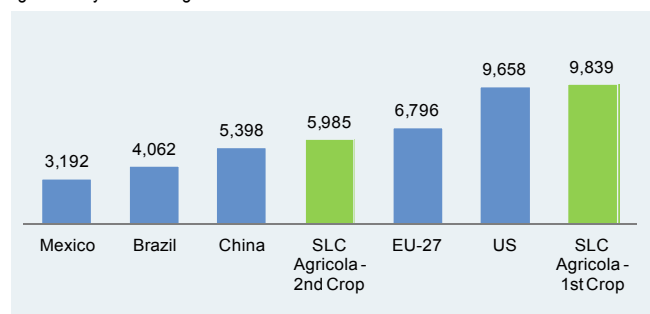
Kg/ha – 5-year average



Source: Company filings.

Figure 3: Global Corn Yields vs. SLCE3

Kg/ha – 5-year average



Source: Company filings.

Rising agricultural land prices to benefit land transformation business

A significant part of SLC's value comes from its land holdings and facilities (modern equipment), and the company currently trades at 0.83x P/NAV, well below its historical average of ~1.1x since 2007 (historical median of 0.9x). In our view, increased agricultural production and planted area expansion should drive land value up in the medium/long term, suggesting that this discount to historical P/NAV is unwarranted. Appreciation in land value also would directly benefit the company's land transformation business.

Finally, the company's recent agreement with Valiance Asset Management to form SLC LandCo, a long-term partnership in order to increase investment in arable land in Brazil, positions the company to continue to invest, unlock value from its existing land holdings and benefit from appreciating land in the long term. With the recent restriction on foreigners owning land in Brazil, SLCE3 provides a vehicle to play this expected long-term land appreciation.

Diversified crop helps mitigate weather volatility and prices pressure

As the company cultivates three main cultures (Soy, Corn and Cotton) alongside its high geographic diversification, this allows SLC Agricola to partially offset downturns in price of its commodities and also mitigate the risk of weather volatility. In addition, the company's crop rotation allows soil characteristics to improve leading to improved yields. SLC has been able to maintain a stable cash margin (average 30% for the last five years) despite the high volatility in grains prices. As a result, even with a more cautious view for soft-commodities prices, we believe the company will be able to keep its EBITDA margin at 30% levels.

We are more cautious on soy and corn prices in the medium term

Though J.P. Morgan forecasts soy and corn prices rebounding in the short term, we see normalization in the medium/long term. The market is coming off of recent highs, which may not allow margin expansion for the company. Our current long-term forecasts for soy, corn and cotton are shown in the table below, and figures including the next three quarters can be found in Appendix I.

Table 3: Grains Price Forecast

Prices (Forecast)	2012	2013	2014	2015
Soy (\$/bu)	15.1	14.3	13.3	12.9
Corn (\$/bu)	7.3	7.6	6.3	6.1
Cotton (\$c/lb)	80.4	77.6	79.6	79.4

Source: Bloomberg and J.P. Morgan.

Current valuation stretched, even including increased yields and land value

Despite the positive outlook for operations, we believe the current valuation leaves limited room for appreciation, especially considering the limited margin growth opportunity that will come with lower prices for soy, corn and cotton.

Company has high sensitivity to commodities prices.

Sensitive to a sudden disruption in commodities prices

Given the company's different crop cycles and earnings recognition, any meaningful changes in commodities prices should lead to volatility in earnings. As commodities prices are highly volatile, we conducted a sensitivity analysis of our estimated valuation to changes in commodities prices. In the following table, we consider all commodities moving up/down by 10% for the long run, and use our long-term commodities price curve for all of them.

Table 4: Price Target – Sensitivity to 10% Change in Long-Term Commodities Curves – 2013E

Commodity	-15%	-10%	0%	10%	15%
Cotton (\$c/lb)	65.96	69.84	77.60	85.36	89.24
Soy (\$/bu)	12.16	12.87	14.30	15.73	16.45
Corn (\$/bu)	6.46	6.84	7.60	8.36	8.74
Adj. EBITDA	202	252	352	452	502
Adj. EBITDA Margin	22%	26%	32%	38%	40%
Price Target (R\$/share)	16.0	19.4	26.1	32.9	36.3

Source: Bloomberg and J.P. Morgan estimates.

Though it is an issue for all agribusiness companies operating in Brazil, it is important to point out that logistics will continue to be a problem.

Brazil infrastructure bottlenecks could weigh on earnings for all Brazilian agricultural players

Logistical bottlenecks are still a primary concern of the agribusiness sector in Brazil. The majority of grain in Brazil is still transported by truck and, until recently, the country's logistical shortcomings were offset by the better yields combined with an increase in planted area. One of the largest discrepancies between Brazil and the US, for instance, relates to transportation costs. In Brazil, to haul a metric ton of soy from the farm to the port, costs are roughly 30% more than those in the US market. Based on data collected from the CTN (National Confederation of Transport), Brazilian railway density is 3.4km/1000 km² vs. 30km/100 km² in the US. In order to continue to compete on a global stage, Brazil needs competitive logistics.

For decades, Brazil's economy has depended heavily on the road transport matrix. We believe this could change; however, we do not believe it will happen in the short to mid-term. The last data released by CTN indicates average investment in Brazil between 1997 and 2000 reached US\$11 thousand/km, while the corresponding US figure was US\$34 thousand/km, according to data released by CNT/COPPEAD. Despite investments, Brazil still has a very low percentage of its roads paved, less than 14% of its total 1.6 million km of roads, which diminishes Brazil's ability to compete with foreign peers. In addition, Brazilian port capacity is far less than that of other major commodity-producing countries.

Risks to Rating and Price Target

Upside risks to our rating and price target include:

- **Commodity price appreciation.** The company's earnings are closely linked to commodities prices, and a 10% increase could provide unexpected upside.
- **Land appreciation.** As investment in Brazil increases, land could appreciate at higher rates than we currently forecast.

Downside risks to our rating and price target include:

- **Global economic slowdown hindering global growth of protein consumption.** As growth in this business is essential for sustaining global grain demand, any reduction in protein consumption could affect adversely SLC's future operating performance.
- **BRL appreciation.** As the company has most of its top line linked to the USD and its costs in BRL, an appreciation of Brazilian Real could pressure the company's margins.
- **Commodity price reduction could lead to earnings volatility.** Given the company's different crop cycles and earnings recognition, any meaningful change in commodity prices could bring volatility to earnings.
- **Delay of infrastructure projects could leave Brazil uncompetitive in the long term.** Costs to haul goods in Brazil are quite expensive when compared to other economies. Though projects are under way to improve infrastructure, downside risks persist.
- **Political risk.** Abrupt changes to land-owning laws, as occurred in 2010, could jeopardize SLC's land transformation business.

Company Description

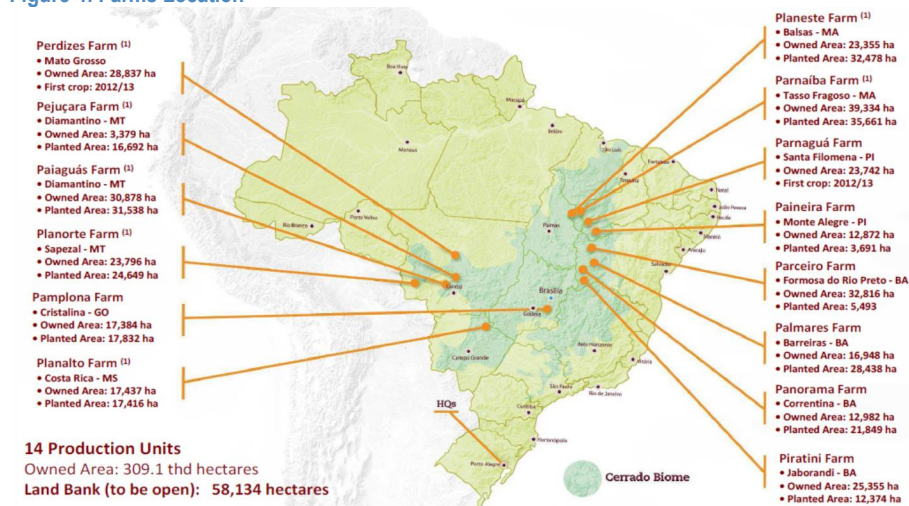
SLC Agricola was founded in mid-1970s. The company is a grain producer focused on soybean, corn, cotton and coffee. The company currently owns 309 thousand hectares of land, of which 248 thousand hectares are planted. In 2010/11, SLC Agricola produced roughly 546 thousand tons of grains and 142 thousand tons of cotton. During the first half of 2012, the company had net revenue of R\$552 million with adjusted EBITDA margin of 32%.

Company Overview

Operations

SLC Agricola operates 14 farms strategically located in Brazil with approximately 309,000 hectares of land, of which 248,000 hectares are planted. The annual share of planted area of these crops in the current crop year is estimated at 38% cotton, 46% soybean, 14% corn and 2% others (coffee mainly). In addition, the company also plants and harvests coffee, although this is a small share of the company's top line. In addition to its traditional farming business, SLC Agricola has a strong track record in the transformation of land in Brazil. In this business, the company usually purchases "raw land" which requires intensive Capex in order to prepare the land to be farm ready, after which the company begins farming the land itself or sells it to interested buyers.

Figure 4: Farms Location



Source: Company; used with permission.

Farming – grains (soybean, corn and coffee) and cotton

SLC Agricola is one of the largest producers of grains in Brazil. In order of cash generation, the company's expected revenue breakdown is 61% cotton, 26% soybean, 10% corn and 3% other crops. The company sells its products to trading companies which subsequently export them. The company's top clients include Cargill, Glencore, Louis Dreyfus, ADM and others.

Table 5: Top-Line Forecast by Commodity – Declining Top-Line Share of Cotton Is Attributed to Increased Planted Area of Soy and Corn, as per Company Guidance and Global Planting Trends
R\$ 000

	2012E	2013E	2014E	2015E
Top Line	1,193	1,444	1,434	1,558
Cotton	590	412	485	574
Soybean	294	382	354	390
Corn	131	258	209	190
Coffee	12	13	31	13
Others	13	27	29	30
Bio. Asset	152	352	327	360

Source: J.P. Morgan.

Land transformation

In addition to its farming business, the company has a land transformation business, the primary objective of which is to search for and buy farmland with high potential for price appreciation. According to company filings, from 1980 to 2012 SLC Agricola invested nearly US\$419 million in land purchases, which today are valued by an independent consultant at market value of US\$1,200 million, yielding an IRR close to 15%.

Land acquisition and leasing for foreigners in Brazil – a shelter in favor of SLC

There has been a lot of noise regarding new landowning laws in Brazil over the past two years. To clarify, the original 5.709 from 1971 does not forbid foreign investors from owning land. However, this was recently overridden by an addendum to the law which restricts the amount of land that foreigners can own. These restrictions are based on (i) 25% of the area per municipality; (ii) 40% (out of the 25%) per foreign country and (iii) the municipality's Registry Office being responsible for this control. Because it is harder for foreign investors to buy land, landowning companies like SLC have a built-in competitive advantage. Furthermore, shares of SLC become a vehicle through which some foreign-based investors can indirectly "own" land in the country.

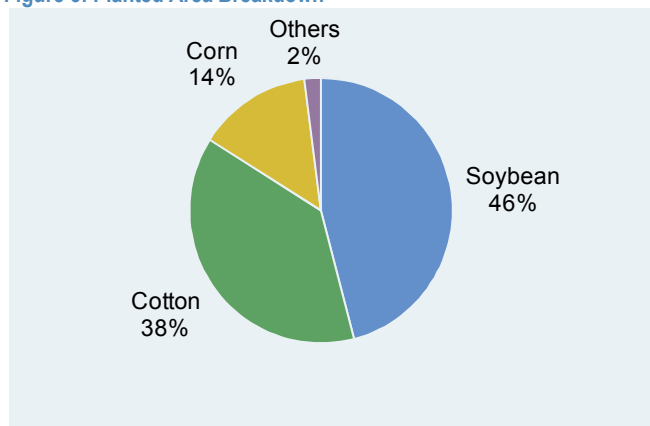
SLC Landco

Recently, SCL LandCo entered into an agreement with Valiance Asset Management in order to develop a long-term partnership in order to increase investment in arable land in Brazil. Valiance will make a capital injection of nearly US\$240 million in the joint venture. SLC will contribute with ~15% of its current land holdings. The entire joint venture stems from the new limits on foreign ownership of land, and we view the concept as favorable for SLC as it allows the company to "unlock" the value inherent in its significant land holdings.

Production process

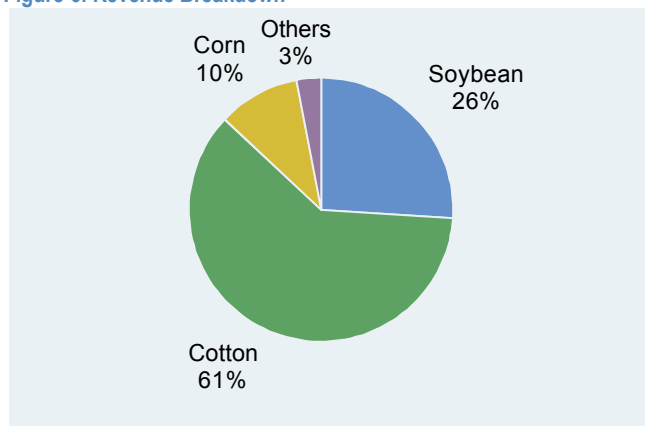
Based on ideal Brazilian climate conditions and also its strategy of crop rotation, the company is able to adjust its production according to what it expects for commodity prices (futures market). This flexibility is a competitive advantage for the company. The steps listed in the below chart include: Agricultural Strategy, Input Purchases, Soil Preparation, Harvesting, Benefitting and Storage, and Shipping.

Figure 5: Planted Area Breakdown



Source: Company.

Figure 6: Revenue Breakdown



Source: Company.

Figure 7: Production Process



Source: Company; used with permission.

In addition, the company also listed a few key proficiencies essential to maintaining its level of flexibility/productivity (yields) which should in turn enhance profitability: (i) soil and climate conditions, (ii) professional management; (iii) high-tech machinery, (iv) research field, and (v) no-till technique. Major investments in technology and capex in the land development process allow the company to have one of the best yields compared with the average in Latin America and also the US.

Cost breakdown

As for most manufacturers of soft commodities, variable costs account for a substantial portion of total costs. The average breakdown between variable and fixed costs stands at 80% (VC) to 20% (FC).

Among variable costs, fertilizers and defensives represent the biggest investments to deliver higher yields. These products account for roughly 50% of the company's

variable costs. In addition, as most of these products are linked to the U.S. dollar, the company usually has to pay upfront in order to guarantee the supply on time, which requires the company to be disciplined in its use of working capital.

For fertilizers/defensives costs, we use forecasts from J.P. Morgan Chemicals team.

Table 6: Fertilizers and Defensives Price Forecast

US\$/ton

	12/13E	13/14E	14/15E	15/16E
Nitrogen	226	226	158	158
Phosphate	623	593	534	534
Potash	601	601	501	501

Source: J.P. Morgan Chemicals Team.

On the fixed-cost side, the largest portion represents labor which accounts for nearly 20% of the company's fixed costs.

Financial Outlook

The main assumptions in our top-line forecast are based on (i) our global commodities long-term price curve and (ii) supply-side inputs from operations of farming business division (production and yield). The table below shows the assumptions we use for SLC's farming business.

Table 7: Commodities Yield Forecast

Yields (Forecast)	2012	2013	2014	2015
Cotton (t/ha)	1.62	1.63	1.66	1.68
Soy (t/ha)	3.30	3.33	3.37	3.40
Corn (t/ha)	7.62	7.67	7.63	7.33

Source: Company data and J.P. Morgan estimates.

Table 8: Commodities Volume Forecast – SLC Guidance

Production Volumes (kt)	2012	2013	2014	2015
Cotton	155	127	168	180
Soy	386	493	486	539
Corn	244	374	359	328

Source: Company data and J.P. Morgan estimates.

Table 9: Top Line – Assumptions

	2012E	2013E	2014E	2015E
Top Line	1,193	1,444	1,434	1,558
Cotton	590	412	485	574
Soybean	294	382	354	390
Corn	131	258	209	190
Coffee	12	13	31	13
Others	13	27	29	30
Bio. Asset	152	352	327	360

Source: J.P. Morgan estimates.

Valuation

We value SLC using an equal blend between DCF from the farming operations and the NAV of its marked-to-market assets (land, machinery, etc.). We do not apply any valuation discount to the land business; however, we include a sensitivity table to show such variation. We use an equal weighting for these two valuation metrics because we believe these are the most common metrics used by investors to value the company today. Furthermore, it appropriately accounts for the interrelationship between the two businesses, i.e., balancing the cash flows expected from the farming business (DCF) with Net Asset Value (NAV).

Table 10: NAV – Land and Equipment Valuation

	R\$M
Land	2052
Machinery	584
Storage Capacity	96
Mills (Cottob, Coffee)	41
Biological Assets	380
EV	3153
Net Debt	732
Implied Market Cap	2421
Shares (mm)	99
Implied Price	24.5
Current Price	19.9
P/NAV (x)	0.81

Source: J.P. Morgan estimates.

Farming business

Our DCF analysis arrives at a valuation slightly less than that of valuing the land separately. Nevertheless, we believe blending these two valuation methods is the most appropriate strategy, given the dependent relationship between the land and farming businesses. In addition, we include an analysis of how the farming business could be valued if SLC operated without its land business.

Table 11: FCFF

	2013E	2014E	2015E	2016E	2017E	2018E	Perp.
EBIT	191	239	272	276	297	324	341
Taxes	(39)	(49)	(56)	(56)	(61)	(66)	(116)
Depreciation	105	107	109	111	113	114	77
Cash Earnings to Firm	256	297	325	330	350	372	302
Chg WK	40	(28)	(34)	(27)	(19)	(30)	15
Capex	(174)	(156)	(159)	(162)	(137)	(137)	(77)
FCFF	122	114	133	141	194	205	4,245

Source: J.P. Morgan estimates.

Table 12: DCF

(+) Present Value of Cash Flows	576
(+) Present Value of Perpetuity	2,618
(=) Present Value of CF to Firm	3,194
(-) Net Debt	445
(-) Minority Interest	0
(=) Equity Value	2,749
(/) # of Shares	99
(=) Equity Value US\$/sh	27.80

Source: J.P. Morgan estimates.

Table 13: WACC

Cost of Equity:	13.4%
Beta	1.0
Risk Free Rate	3.5%
Country Risk	2.2%
Equity Risk Premium	5.5%
Cost of Debt:	4.1%
% Equity	65.0%
% Debt	35.0%
WACC	10.2%

Source: J.P. Morgan estimates.

As a result, we arrive at Dec-13 price target of R\$26, as detailed in the table below.

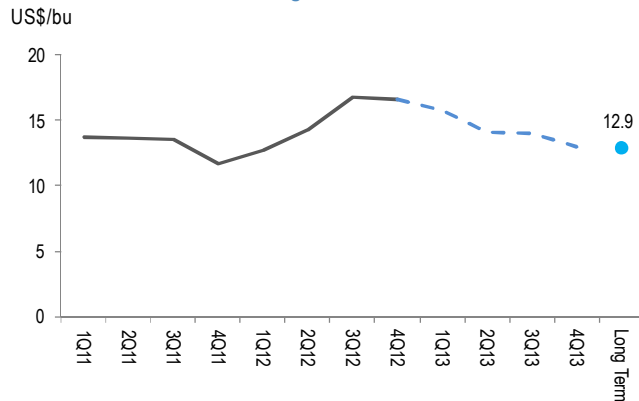
Table 14: Blended Price Target

	R\$/sh	Mult.	Weight	% Upside
DCF	28	10.2% WACC	50%	32%
Replacement Value	24		50%	17%
Price Target	26		100%	24%
Total Return				24%

Source: J.P. Morgan estimates.

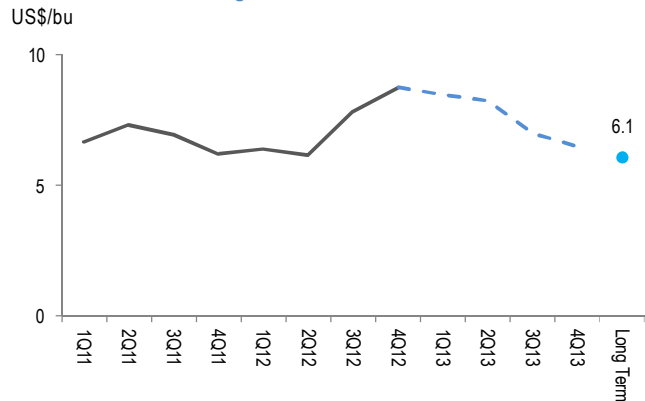
Appendix I: Price Outlook for Key Commodities

Figure 8: Soy – J.P. Morgan Forecasts Soy Prices Rebounding in the Short Term Before Normalizing



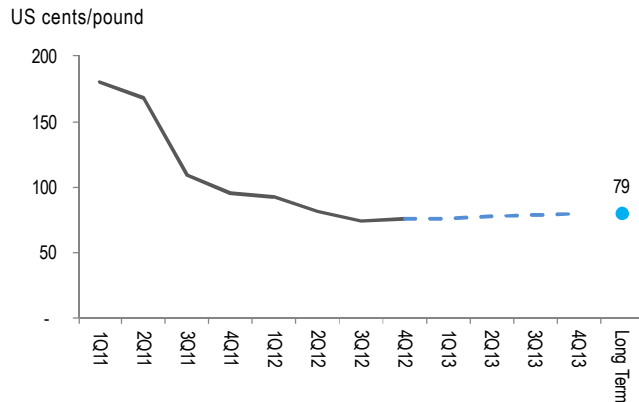
Source: Bloomberg, J.P. Morgan estimates.

Figure 9: Corn – J.P. Morgan Forecasts a Similar Recovery in Corn Before a Trend Lower Again



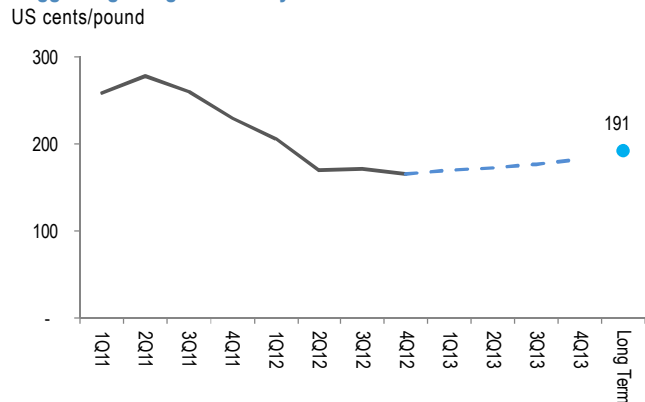
Source: Bloomberg, J.P. Morgan estimates.

Figure 10: Cotton – Off from Historical Highs and We Currently Forecast Price Stabilization



Source: Bloomberg, J.P. Morgan estimates.

Figure 11: Coffee – We Follow the Forward Curves of the Market, Suggesting a Slight Recovery from Current Levels



Source: Bloomberg, J.P. Morgan estimates.

Appendix II: Overview of Grains and Cotton

Supply-Side Analysis Should Drive Investment Decisions

We expect demand to continue growing and thus believe supply-side analysis should drive investment decisions.

Over the next decade, agricultural demand will be driven by (i) population growth, (ii) rural to urban migration, (iii) changing consumer tastes in the developed world and (iv) new uses for current crops. Grain supply will be driven by any increase in planted area and the evolution of crop yields. For this reason, supply-side factors such as yield improvement and greater efficiencies are likely to be the differentiating factors for agribusiness companies like SLC Agricola.

Grains – Driven by Planted Area and Crop Yields

We view the primary drivers for supply of soy and corn as the outlook for planted area and yields. Demand for grains is expected to increase thanks to continued growth in (i) population, (ii) urbanization rates and (iii) meat consumption, particularly in developing economies, over the coming decades. Increased industrial uses for grains will also broaden demand for certain crops, particularly corn given its use to produce ethanol and sweeteners.

Global grain supplies will be driven by planted area and crop yields

Planted area and crop yields will drive supply in Brazil moving forward.

In our view, supply will be the driver of the global grains balance and thus commodity prices. We identify the two primary drivers of grain supply as planted area and crop yields. Planted area, determined by a farm's crop allocation, is affected by (i) expected profitability of each crop, (ii) available land and (iii) subsidies from the government, among other factors. Yield, on the other hand, is determined largely by (i) weather conditions, (ii) technology applied to the soil and/or crops and (iii) disease.

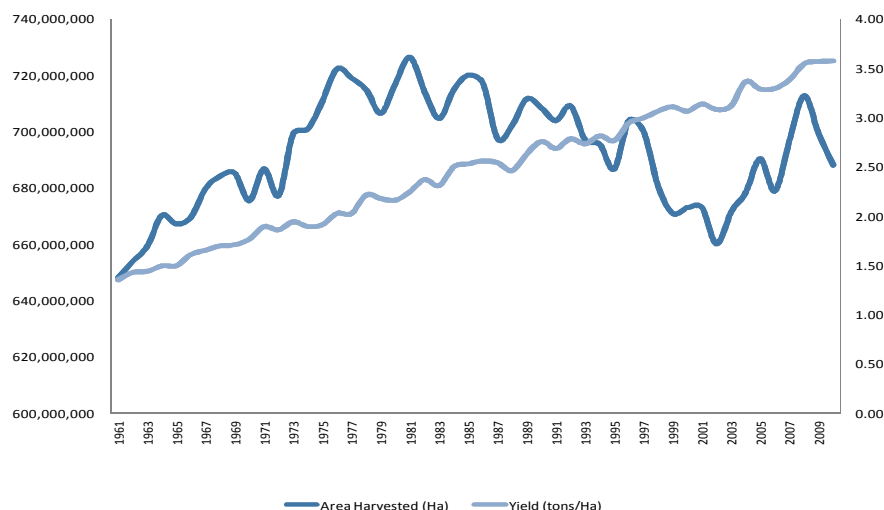
Growth in planted area should slow in the coming years.

According to the OECD-FAO, for the last 40 years, the planted area in the world has grown at a timid 0.6%, while arable land expanded at a slightly better 0.8% per year (CAGR 1961-2009). At the same time, yields have improved due to the high level of investment in technology and the development of new irrigation processes, machinery and modified strains of crops. We do not see any supply shocks for soy or corn in the near future, and forecast the supply to continue to grow at a moderate pace in the medium term. More pressing would be slowing growth in arable land (there is a fixed amount on earth), less improvement in yields and a boom in demand.

Meanwhile, yields should continue to improve, especially in Brazil where there is still upside from investment in technology.

During the last 50 years, the planted area oriented to grains globally has been nearly flat. At the same time, yields for cereals have been improving at approximately 2% per annum (CAGR 1961-2010), largely attributed to new technologies which allow producers to generate more cash from their existing land.

Figure 12: Yield vs. Area Harvested – Globally



Source: OECD-FAO.

Improving yields comes down to technology

Weather conditions and investment in technology are the factors that have affected crop yields most in the past 50 years. Weather is the most crucial factor for any crop, and it cannot be controlled. However, companies can actively invest in technology, which gives large-scale producers with resources a significant advantage on the global stage.

While weather cannot be controlled, companies can improve yields by investing in technology.

Weather conditions represent one of the most important variables and Brazil has a substantial advantage. Fluctuations in weather bring structural uncertainties into grain harvests and substantially impact prices. In addition to seasonal variations in rainfall and sunshine, J.P. Morgan's Soft Commodities Team identifies water constraints and global warming as two larger issues related to weather that could affect crops in the long term. With a finite quantity of freshwater in the world, it is likely that certain regions stress their supplies as they try to irrigate increasingly large areas of farmland. Climate change or global warming could have the effect of reducing crop output, affecting crop yields or potentially shifting the geographical location of the most important farming regions of the world.

Key weather issues include global water constraints and the secondary effects of climate change.

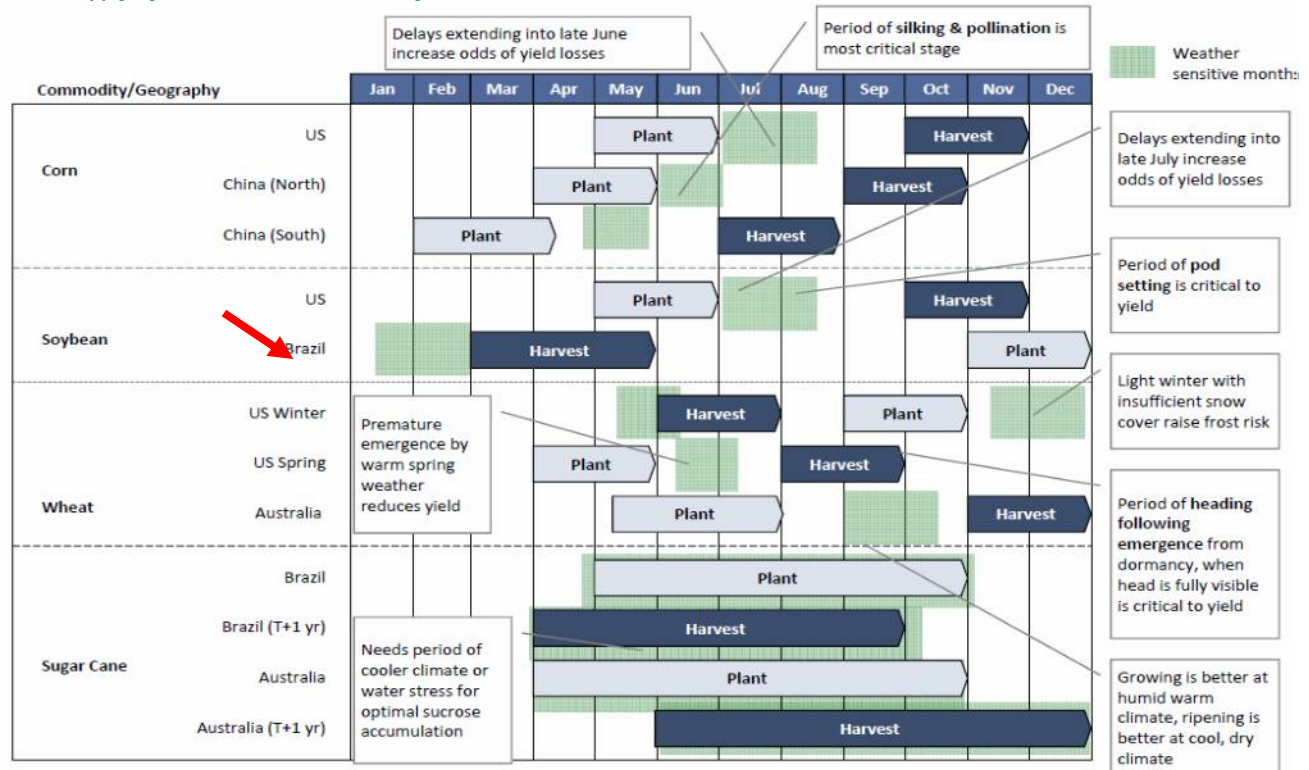
Compared to other regions such as the US (especially the Corn Belt), the average year temperature in Brazil has historically exhibited less variation than the global average, allowing the country to have two harvests in the same calendar year. This is called the *safrinha*, and has positive implications for corn and cotton. These climate conditions allow Brazil to have one of the best production yields globally, with significant room for upside given the historical lack of investment in agricultural technologies, which producers only began to implement decades ago, through substantial and deep studies from EMBRAPA, the Brazilian Agricultural Research Corporation.

Brazil has a comparative advantage with stable weather relative to other major agricultural areas.

Because Brazil's crop cycles differ from those in the Northern Hemisphere, it is able to serve as a "global supply shelter," normalizing price and end-stock levels between Northern Hemisphere harvests. In the chart below, from our Global Commodities Team, one can see Brazil's crop cycle compared with those in the North Hemisphere.

Being in the Southern Hemisphere, Brazil is able to serve as a global supply shelter for many crops.

Figure 13: Supply Cycles and Weather Sensitivity



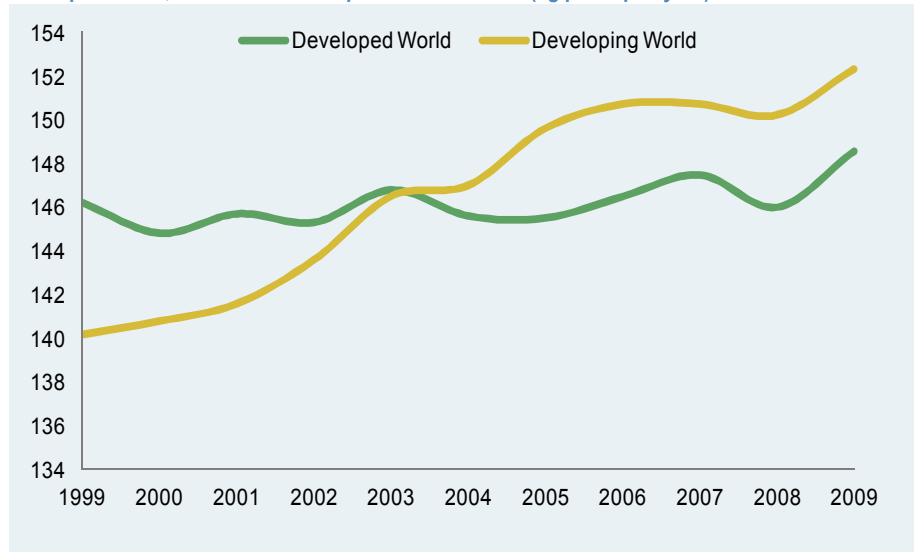
Source: J.P. Morgan Global Commodities Team.

Global grains demand will keep rising

Population growth, rural to urban migration, higher per-capita income and new industrial uses will drive grains demand growth.

We expect grains demand to continue to grow at consistent rates. According to the OECD-FAO, agricultural production needs to increase by 60% over the next 40 years to meet rising demand for food, not including the potential increase in the use of corn-based ethanol. Soy and corn are primarily used for animal feed, human consumption and industrial uses. Consumption of grains has increased at a compound annual rate of 0.8% in the last 50 years. This growth is expected to continue, driven by increased demand in emerging economies and expanded industrial uses in developed ones. Moving forward, we view the primary drivers of global grains demand as (i) world population growth, (ii) rural to urban migration, (iii) higher per-capita incomes which lead to increased meat consumption and (iv) new industrial demand for certain crops.

Figure 14: Grains Consumption in the Developing World Recently Surpassed Consumption in the Developed World, a Trend That Is Expected to Continue (kg/per capita/year)



Source: FAOSTAT.

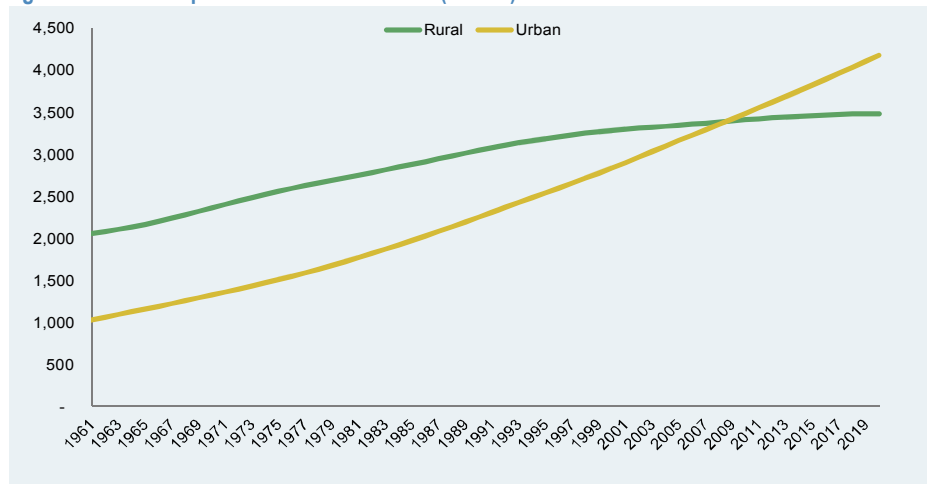
A growing population naturally elevates demand for grains.

Population growth will directly increase demand for grains. According to OECD-FAO, grains production should have to double from its current levels in order to meet demand for the next three decades.

This process is accelerated with rural to urban migration in developing countries.

Rural to urban migration. The global migration from rural to urban areas should accelerate demand for efficient, large-scale agriculture, including grains, as higher concentrations of people living together will need to be fed. This migration also increases the need for efficiency, as a larger urban population must be fed by a dwindling rural community.

Figure 15: World Population – Rural vs. Urban (million)



Source: OECD-FAO.

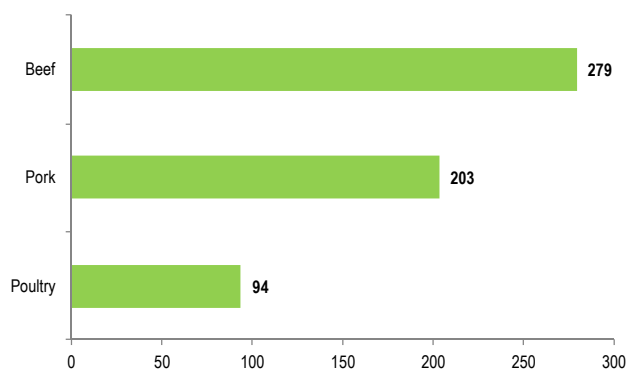
Increased living standards lead to increased demand for meat, as families can include more meat in their diets as their wealth improves.

Increased living standards should boost global meat consumption, further increasing grain demand. As living standards improve globally, populations are expected to begin to consume more meat. This trend is linked to population growth, rural to urban migration and general economic expansion: with urban migration comes higher wages and changing consumption patterns, with many families able to afford more meat in their diets for the first time. Growth in meat consumption should accelerate especially in developing economies in Asia and Africa, as there is significant upside to consumption of meat in these areas compared to the developed world.

The feed conversions show the increased stress on grain supply from meat consumption.

A key point here is that increased meat consumption increases exponentially the amount of grains consumed by a population. Poultry, pork and beef have increasing feed conversions, or the amount of grains needed to produce a kg of meat. A recent investor presentation from Tyson estimates the relationship and illustrates how the growth in meat consumption creates significant demand for grains.

Figure 16: Kg of Corn Required to Produce 45 kg (100 lbs) of Meat



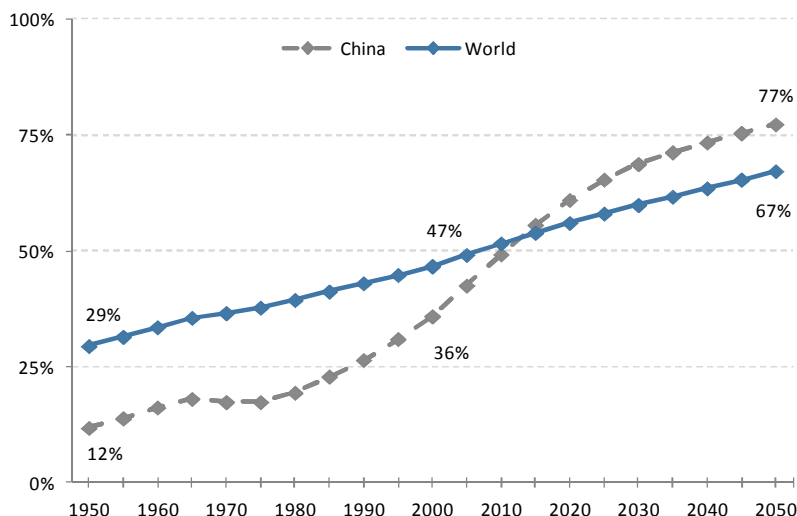
Source: Tyson Investor Presentation, J.P. Morgan Global Protein Conference, March 2011.

- **For an independent beef feedlot**, ~279 kg of corn is needed to produce 45 kg of boneless fed beef (11 bushels to produce 100 lbs).
- **For an independent producer of pork**, ~203 kg of corn is needed to produce 45 kg of boneless skinless meat (8 bushels to produce 100 lbs).
- **For a contract grower of poultry**, ~94 kg of corn is needed to produce 45 kg of boneless skinless meat (3.7 bushels to produce 100 lbs).

China embodies these three growth dynamics and should continue to drive global demand.

China should continue to be the largest grains consumer for the next decade as its necessity to feed around 1.4 billion people should continue to expand. The combination of available income and migration from rural to urban areas should strongly contribute to this as well. As a result, consumption of meat and more proteins should increase. China, given its preference for pork, is by far the largest protein consumer in the world, and this volume is expected to continue growing as the country's per-capita level is still well below the average in developed economies.

Figure 17: China's Urbanization Rate Is Expected to Pass the World Average in the Next Decade
Urbanization Rate %



Source: United Nations, 2011.

Meat consumption should be led by poultry over the next decade.

Meat consumption moving forward. Figures released by the OECD-FAO and the USDA suggest pork, beef and poultry production should continue growing for the next 10 years, with beef outperforming the other meats in developed countries. Poultry should continue to grow strongly, mainly in the developing economies where consumption over the next 10 years is expected to grow at a compound annual rate of 2.8%, compared to growth for pork at 2.1% and beef at 1.8%.

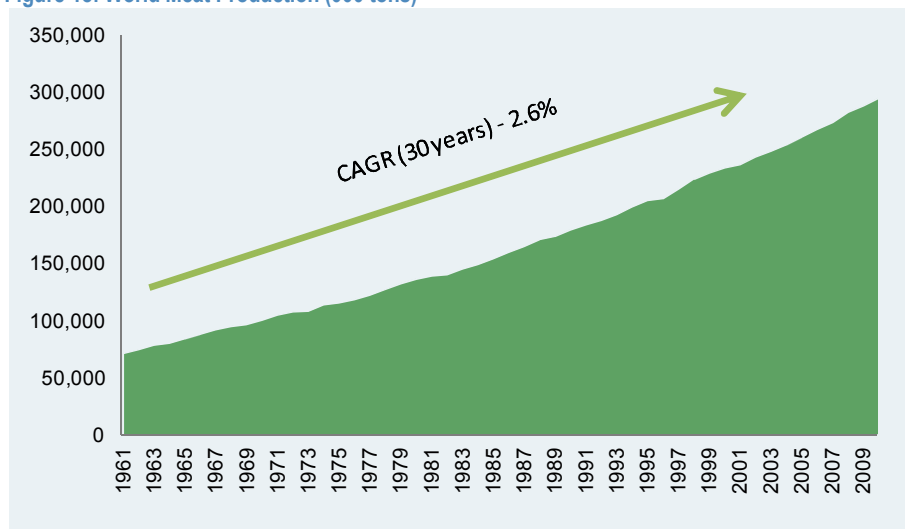
Table 15: Meat Consumption Growth (Past and Future)

	2001-2010	2011-2020	2001-2010	2011-2020
	Developed World		Developing World	
Poultry	2.07%	1.51%	4.3%	2.8%
Beef	0.05%	1.05%	2.3%	1.8%
Pork	1.15%	0.82%	2.1%	2.1%

Source: FAOTSTAT.

For the past 30 years meat production has grown at a compound annual rate of 2%. As feedlots have become more common, they have greatly improved the yield per hectare for meat producers. Instead of having to allocate 1 hectare per head of cattle, producers can increase efficiency using feedlots of soy and corn.

Figure 18: World Meat Production (000 tons)



Source: FAOSTAT.

Biofuel will also increase demand for corn used in US corn ethanol.

Increased use of biofuel will constitute an important component of grains demand in the next decade. In order to accelerate the reduction of greenhouse gas emissions, demand for biofuel in the US should continue growing with demand for new green vehicles. According to the RFA (Renewable Fuel Association), by 2015 the US is expected to consume roughly 20 billion gallons per annum of biofuel (renewable and advanced), which should increase demand for corn. The table below shows the US schedule for Renewable Biofuel which, we believe, should drive demand for corn. This demand could be partially offset by the introduction of new technology applied in modern refineries that could increase existing productivity levels.

Since 1999, the number of refineries in the US has grown at 12% CAGR and there are now nearly 209 refineries with capacity to produce nearly 15 billion of gallons per year, up from 1.7 billion 13 years ago.

Table 16: Renewable Fuel Standards Timeline (billion gallons/year)

Year	2009	2010	2011	2012	2013e	2014e	2015e	2016e	2017e	2018e	2019e	2020e	2021e	2022e
Renewable Biofuel	10.5	12.0	12.6	13.2	13.8	14.4	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
Advanced Biofuel	0.6	1.0	1.4	2.0	2.8	3.8	5.5	7.3	9.0	11.0	13.0	15.0	18.0	21.0
Cellulosic Biofuel	0.0	0.1	0.3	0.5	1.0	1.8	3.0	4.3	5.5	7.0	8.5	10.5	13.5	16.0
Biomass-based Diesel	0.5	0.7	0.8	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Undifferentiated Advanced Biofuel	0.1	0.2	0.3	0.5	1.8	2.0	2.5	3.0	3.5	4.0	4.5	4.5	4.5	5.0
Total RFS	11.1	13.0	14.0	15.2	16.6	18.2	20.5	22.3	24.0	26.0	28.0	30.0	33.0	36.0

Source: RFA – Renewable Fuel Association.

Though our commodities team forecasts a short-term rebound in corn and soy prices, prices should trend downwards in the medium/long term, in line with historical levels.

Grains Price

Corn and soy price levels are currently high as demand has spiked from China while supply has not followed the same path. In addition, production costs have increased over the last decades, also contributing to increased prices. The table below shows our price forecast for the commodities grains.

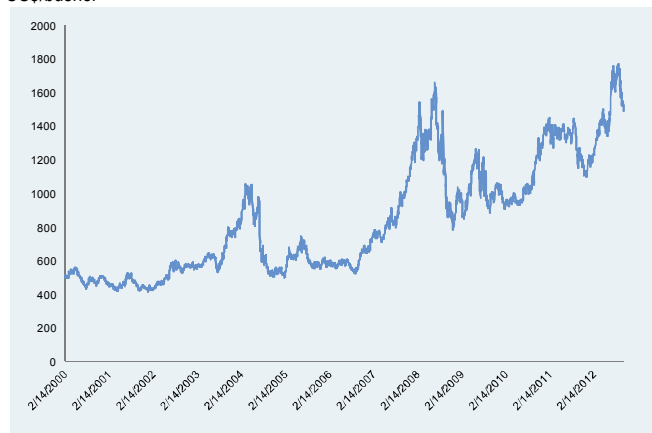
Table 17: Grains Price Forecast

Prices (Forecast)	2012	2013	2014	2015
Soy (\$/bu)	15.1	14.3	13.3	12.9
Corn (\$/bu)	7.3	7.6	6.3	6.1

Source: Bloomberg, J.P. Morgan estimates.

Figure 19: Soy Price Performance Since 2000 (10-year average of US\$895/bushel)

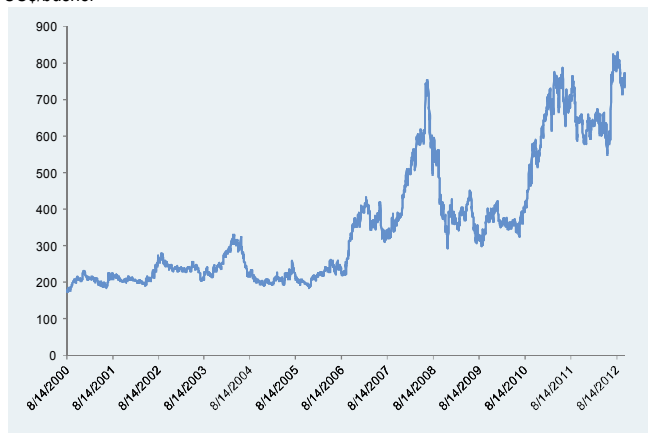
US\$/bushel



Source: Bloomberg.

Figure 20: Corn Price Performance Since 2000 (10-year average of US\$393/bushel)

US\$/bushel



Source: Bloomberg.

Growth in cotton demand should continue, but at a slower rate than in past years.

Cotton – Off Historical Peak

Cotton is used globally for various types of apparel, sheets and other products. While we expect demand for grains to continue to grow strongly in the coming decade, we think cotton could post less substantial growth, especially as technology enables the introduction of new competitive fabrics in the market. According to the last *Cotton and Wool Outlook* (USDA), cotton production is expected to decline to 116.3 million bales in 2012/13, or 6% below the 124 million produced in the previous year. While production is expected to experience a slight drop, consumption is forecast to advance by 3.6% to nearly 107 million bales.

Table 18: Global Cotton Production (million bales)

06/07	07/08	08/09	09/10	10/11	11/12	12/13 (Aug)	12/13 (Sept)	12/13 (Oct)
122	120	107	102	116	124	114	114	116

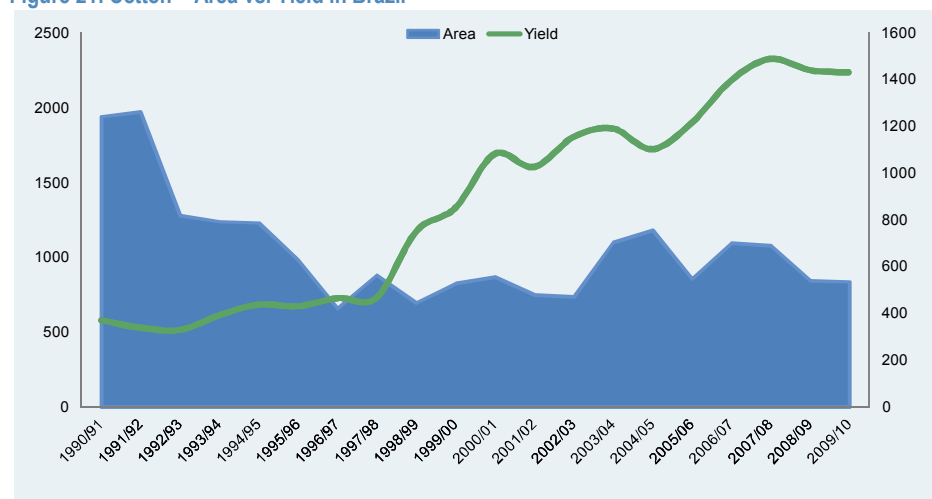
Source: USDA.

Brazil has one of the highest cotton yields in the world.

Cotton in Brazil

Brazil is the fifth-largest cotton producer in the world and accounts for nearly 5% of global cotton output, according to the USDA. The remaining four large producers are the US, China, India and Pakistan. The US, India and China provide approximately 66% of the world's cotton. The US is the leading exporter and accounts for nearly 1/3 of total raw cotton trade. Eighty percent of cotton in Brazil is harvested in the regions of Mato Grosso and Bahia. During the past decades, Brazil has adopted modern technology for cotton planting and harvesting, enabling the country to reach one of the highest yields surpassing by far the world average.

Figure 21: Cotton – Area vs. Yield in Brazil



Source: CONAB (Companhia Nacional de Abastecimento).

Figure 22: Cotton Price Reached Historical Highs in 2010 (10-year average of 70.2 cents/lb)
US cents/lb



Source: Bloomberg (ICE).

Cotton price outlook

Prices of cotton should stay at higher-than-average levels as competition with other grains and oilseeds intensifies. Our perpetuity price assumption stands at US\$0.79/lb.

Table 19: Cotton Price Forecast

Prices (Forecast)	2012	2013	2014	2015
Cotton (\$c/lb)	80.39	77.60	79.57	79.36

Source: J.P. Morgan estimates.

Global outlook for cotton supply and demand

- **Global 2012/13 cotton production forecast to decline 6%.** According to the USDA, the rising prices of competing crops and the declining price of fiber is rendering cotton cultivation less profitable globally.
- **The decline in production is particularly strong in Brazil and China for the coming crop year.** Brazil production is expected to drop 25% yoy, with lowered demand for cotton and the still-strong prices of soybeans expected to shift planted area away from cotton. The USDA currently forecasts a drop of 29% yoy. Cotton production in China is expected to fall 5% yoy. Central government support for grains and the rising cost of cotton production are contributing to this decline.
- **In the long term, global water scarcity could raise the cost profile of cotton relative to other crops.** Fresh water is becoming scarcer globally, and growing cotton demands roughly 3% of the world's water use. We believe this long-term trend could push farmers to cultivate crops with lower water needs.
- **Substitution of goods will be another long-term trend to be watched.** As cotton is widely used in apparel production, many textile companies also have started to use synthetic fibers as a substitute for cotton. As these technologies and their affordability improve, they will continue to present an important substitute for cotton for traditional use in fabrics.

SLC Agricola: Summary of Financials

Income Statement	FY11A	FY12E	FY13E	FY14E	FY15E	Balance Sheet	FY11A	FY12E	FY13E	FY14E	FY15E
Revenues	1,006	1,192	1,444	1,434	1,558	Cash	130	90	226	313	417
Cost of goods sold	433	874	1,042	981	1,062	Accounts receivable	175	149	157	159	172
SG&A	111	107	106	107	114	Inventories	410	242	259	275	292
Depreciation	69	95	105	107	109	Other current assets	7	7	7	7	7
EBITDA	464	211	295	346	381	Net PP&E	2,549	2,615	2,684	2,733	2,782
EBITDA margin	46.1%	17.7%	20.5%	24.1%	24.5%	Other assets	7	7	7	7	7
Financial income	47	19	1	2	3	Total assets	3,648	3,517	3,692	3,862	4,057
Financial expense	(73)	(52)	(7)	(7)	(8)	Short-term debt	385	260	260	278	295
FX & Monetary gains (losses)	10	(21)	6	(7)	(6)	Accounts payable	118	118	126	134	142
Other Nonoperating income	2	1	-	-	-	Other current liabilities	315	301	301	301	301
Equity income	0	0	0	0	0	Long-term debt	255	283	283	302	321
EBT	379	62	193	230	264	Deferred taxes	521	507	507	507	507
Taxes	(81)	(27)	(40)	(47)	(54)	Other liabilities	135	102	102	102	102
Minority interest	0	0	-	-	-	Total liabilities	1,729	1,571	1,580	1,624	1,668
Extraordinary	0	0	-	-	-	Minority interest	0	0	-	-	-
Net income	298	35	154	183	210	Shareholders' equity	1,920	1,946	2,112	2,238	2,390
Net income margin	29.6%	3.0%	10.7%	12.7%	13.5%	Liabilities + Equity	3,649	3,518	3,692	3,862	4,057
EPS	3.01	0.36	1.55	1.85	2.13	Net debt	510	454	317	267	199
Revenue growth	13.2%	18.5%	21.1%	(0.6%)	8.6%	Net Debt/Capital	19.9%	18.2%	11.9%	9.5%	6.6%
EBITDA growth	148.7%	(54.4%)	39.8%	17.1%	10.2%	Debt/Capital	25.0%	21.8%	21.8%	21.8%	21.8%
Net income growth	306.3%	(88.1%)	334.9%	18.8%	15.1%	Net Debt/EBITDA	1.1	2.1	1.1	0.8	0.5
FCF growth	(360.5%)	577.6%	(35.4%)	(45.3%)	26.7%						
Operating Data, Ratios	FY11A	FY12E	FY13E	FY14E	FY15E	Valuation, Macro	FY11A	FY12E	FY13E	FY14E	FY15E
Capex	178	169	174	156	159	EV/EBITDA	5.4	11.7	7.9	6.6	5.8
Change in working capital	(105)	157	40	(28)	(34)	P/E	6.8	57.0	13.1	11.0	9.6
Free cash flow	40	271	175	96	121	P/BV	1.3	1.2	1.2	1.1	1.1
Dividends	0	9	38	46	53	EV/tonne	-	-	-	-	-
Dividend % of net income	0.0%	25.0%	68.0%	58.8%	52.0%	FCF yield	2.0%	13.4%	8.7%	4.7%	6.0%
Capex/depreciation	2.6	1.8	1.7	1.5	1.5	Dividend yield	-	-	-	-	-
CAPEX/sales	17.7%	14.2%	12.1%	10.9%	10.2%	ROE	14.0%	1.6%	6.5%	8.7%	8.6%
Working capital	424	266	226	254	288	Net income margin	29.6%	3.0%	10.7%	12.7%	13.5%
Working capital/sales	42.1%	22.3%	15.7%	17.7%	18.5%	Net revenue/Assets	27.6%	33.9%	39.1%	37.1%	38.4%
						Assets/Equity	1.9	1.8	1.7	1.7	1.7
						ROIC	11.8%	2.6%	5.3%	6.5%	7.2%
						Shares	99	99	99	99	99
						ADRs	99	99	99	99	99

Source: Company reports and J.P. Morgan estimates.

Note: R\$ in millions (except per-share data). Fiscal year ends Dec

Analyst Certification: The research analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple research analysts are primarily responsible for this report, the research analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the research analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect his or her personal views about any and all of the subject securities or issuers; and (2) no part of any of the research analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the research analyst(s) in this report.

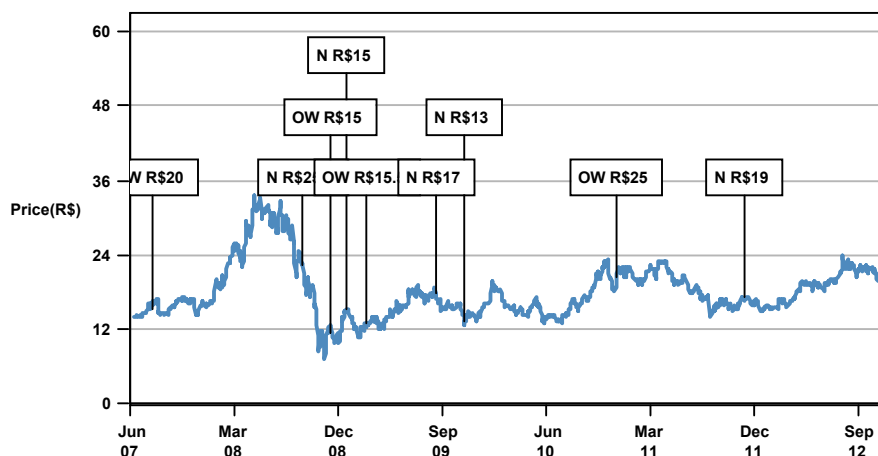
In compliance with Instruction 483 issued by Comissao de Valores Mobiliarios (the Brazilian securities commission) on July 6, 2010, the Brazilian primary analyst signing this report declares: (1) that all the views expressed herein accurately reflect his or her personal views about the securities and issuers; (2) that all recommendations issued by him or her were independently produced, including from the entity in which he or she is an employee; and (3) that he or she will set forth any situation or conflict of interest believed to impact the impartiality of the recommendations herein, as per article 17, II of Instruction 483.

Important Disclosures

- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following company(ies) as clients: SLC Agricola.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following company(ies) as clients, and the services provided were non-investment-banking, securities-related: SLC Agricola.
- **Non-Investment Banking Compensation:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from SLC Agricola.

Company-Specific Disclosures: Important disclosures, including price charts, are available for compendium reports and all J.P. Morgan–covered companies by visiting <https://mm.jpmorgan.com/disclosures/company>, calling 1-800-477-0406, or emailing research.disclosure.inquiries@jpmorgan.com with your request.

SLC Agricola (SLCE3.SA, SLCE3 BZ) Price Chart



Date	Rating	Share Price (R\$)	Price Target (R\$)
27-Jul-07	OW	15.33	20.00
28-Aug-08	N	22.42	25.70
07-Nov-08	OW	11.41	15.00
22-Dec-08	N	15.18	15.00
13-Feb-09	OW	12.89	15.50
13-Aug-09	N	17.80	17.00
27-Oct-09	N	13.30	13.00
02-Dec-10	OW	20.34	25.00
01-Nov-11	N	17.08	19.00

Source: Bloomberg and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jul 27, 2007.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period.

J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight [Over the next six to twelve months, we expect this stock will outperform the average total return of the stocks in the analyst's (or the analyst's team's) coverage universe.] Neutral [Over the next six to twelve months, we expect this stock will perform in line with the average total return of the stocks in the analyst's (or the analyst's team's) coverage universe.] Underweight [Over the next six to twelve months, we expect this stock will underperform the average total return of the stocks in the analyst's (or the analyst's team's) coverage universe.] Not Rated (NR): J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a

recommendation or a rating. In our Asia (ex-Australia) and U.K. small- and mid-cap equity research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying analyst's coverage universe can be found on J.P. Morgan's research website, www.morganmarkets.com.

Coverage Universe: Lucin, Cassio: Autometal (AUTM3.SA), Iochpe-Maxion (MYPK3.SA), MAHLE Metal Leve (LEVE3.SA), Marcopolo (POMO4.SA), Mills (MILS3.SA), Randon (RAPT4.SA), Weg (WEGE3.SA)

J.P. Morgan Equity Research Ratings Distribution, as of September 28, 2012

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage	44%	44%	12%
IB clients*	52%	46%	34%
JPMS Equity Research Coverage	42%	48%	10%
IB clients*	69%	61%	53%

*Percentage of investment banking clients in each rating category.

For purposes only of FINRA/NYSE ratings distribution rules, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.morganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com.

Equity Analysts' Compensation: The equity research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of JPMS, are not registered/qualified as research analysts under NASD/NYSE rules, may not be associated persons of JPMS, and may not be subject to FINRA Rule 2711 and NYSE Rule 472 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan ("JPM") is the global brand name for J.P. Morgan Securities LLC ("JPMS") and its affiliates worldwide. J.P. Morgan Cazenove is a marketing name for the U.K. investment banking businesses and EMEA cash equities and equity research businesses of JPMorgan Chase & Co. and its subsidiaries.

Options related research: If the information contained herein regards options related research, such information is available only to persons who have received the proper option risk disclosure documents. For a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options, please contact your J.P. Morgan Representative or visit the OCC's website at <http://www.optionsclearing.com/publications/risks/riskstoc.pdf>

Legal Entities Disclosures

U.S.: JPMS is a member of NYSE, FINRA, SIPC and the NFA. JPMorgan Chase Bank, N.A. is a member of FDIC and is authorized and regulated in the UK by the Financial Services Authority. **U.K.:** J.P. Morgan Securities plc (JPMS plc) is a member of the London Stock Exchange and is authorized and regulated by the Financial Services Authority. Registered in England & Wales No. 2711006. Registered Office 25 Bank Street, London, E14 5JP. **South Africa:** J.P. Morgan Equities Limited is a member of the Johannesburg Securities Exchange and is regulated by the FSB. **Hong Kong:** J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong. **Korea:** J.P. Morgan Securities (Far East) Ltd, Seoul Branch, is regulated by the Korea Financial Supervisory Service. **Australia:** J.P. Morgan Australia Limited (ABN 52 002 888 011/AFS Licence No: 238188) is regulated by ASIC and J.P. Morgan Securities Australia Limited (ABN 61 003 245 234/AFS Licence No: 238066) is a Market Participant with the ASX and regulated by ASIC. **Taiwan:** J.P.Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. **India:** J.P. Morgan India Private Limited, having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz East, Mumbai - 400098, is a member of the National Stock Exchange of India Limited (SEBI Registration Number - INB 230675231/INF 230675231/INE 230675231) and Bombay Stock Exchange Limited (SEBI Registration Number - INB 010675237/INF 010675237) and is regulated by Securities and Exchange Board of India. **Thailand:** JPMorgan Securities (Thailand) Limited is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. **Indonesia:** PT J.P. Morgan Securities Indonesia is a member of the Indonesia Stock Exchange and is regulated by the BAPEPAM LK. **Philippines:** J.P. Morgan Securities Philippines Inc. is a member of the Philippine Stock Exchange and is regulated by the Securities and Exchange Commission. **Brazil:** Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. **Mexico:** J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is a member of the Mexican Stock Exchange and authorized to act as a broker dealer by the National Banking and Securities Exchange Commission. **Singapore:** This material is issued and distributed in Singapore by J.P. Morgan Securities Singapore Private Limited (JPMS) [MICA (P) 088/04/2012 and Co. Reg. No.: 199405335R] which is a member of the Singapore Exchange Securities Trading Limited and is regulated by the Monetary Authority of Singapore (MAS) and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore) which is regulated by the MAS. **Malaysia:** This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X) which is a Participating Organization of Bursa Malaysia Berhad and a holder of Capital Markets Services License issued by the Securities Commission in Malaysia. **Pakistan:** J. P. Morgan Pakistan Broking (Pvt.) Ltd is a member of the Karachi Stock Exchange and regulated by the Securities

and Exchange Commission of Pakistan. **Saudi Arabia:** J.P. Morgan Saudi Arabia Ltd. is authorized by the Capital Market Authority of the Kingdom of Saudi Arabia (CMA) to carry out dealing as an agent, arranging, advising and custody, with respect to securities business under licence number 35-07079 and its registered address is at 8th Floor, Al-Faisaliyah Tower, King Fahad Road, P.O. Box 51907, Riyadh 11553, Kingdom of Saudi Arabia. **Dubai:** JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - Building 3, Level 7, PO Box 506551, Dubai, UAE.

Country and Region Specific Disclosures

U.K. and European Economic Area (EEA): Unless specified to the contrary, issued and approved for distribution in the U.K. and the EEA by JPMS plc. Investment research issued by JPMS plc has been prepared in accordance with JPMS plc's policies for managing conflicts of interest arising as a result of publication and distribution of investment research. Many European regulators require a firm to establish, implement and maintain such a policy. This report has been issued in the U.K. only to persons of a kind described in Article 19 (5), 38, 47 and 49 of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (all such persons being referred to as "relevant persons"). This document must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is only available to relevant persons and will be engaged in only with relevant persons. In other EEA countries, the report has been issued to persons regarded as professional investors (or equivalent) in their home jurisdiction. **Australia:** This material is issued and distributed by JPMSAL in Australia to "wholesale clients" only. JPMSAL does not issue or distribute this material to "retail clients". The recipient of this material must not distribute it to any third party or outside Australia without the prior written consent of JPMSAL. For the purposes of this paragraph the terms "wholesale client" and "retail client" have the meanings given to them in section 761G of the Corporations Act 2001. **Germany:** This material is distributed in Germany by J.P. Morgan Securities plc, Frankfurt Branch and J.P.Morgan Chase Bank, N.A., Frankfurt Branch which are regulated by the Bundesanstalt für Finanzdienstleistungsaufsicht. **Hong Kong:** The 1% ownership disclosure as of the previous month end satisfies the requirements under Paragraph 16.5(a) of the Hong Kong Code of Conduct for Persons Licensed by or Registered with the Securities and Futures Commission. (For research published within the first ten days of the month, the disclosure may be based on the month end data from two months prior.) J.P. Morgan Broking (Hong Kong) Limited is the liquidity provider/market maker for derivative warrants, callable bull bear contracts and stock options listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>. **Japan:** There is a risk that a loss may occur due to a change in the price of the shares in the case of share trading, and that a loss may occur due to the exchange rate in the case of foreign share trading. In the case of share trading, JPMorgan Securities Japan Co., Ltd., will be receiving a brokerage fee and consumption tax (shouhizei) calculated by multiplying the executed price by the commission rate which was individually agreed between JPMorgan Securities Japan Co., Ltd., and the customer in advance. Financial Instruments Firms: JPMorgan Securities Japan Co., Ltd., Kanto Local Finance Bureau (kinsho) No. 82 Participating Association / Japan Securities Dealers Association, The Financial Futures Association of Japan, Type II Financial Instruments Firms Association and Japan Investment Advisers Association. **Korea:** This report may have been edited or contributed to from time to time by affiliates of J.P. Morgan Securities (Far East) Ltd, Seoul Branch. **Singapore:** JPMS and/or its affiliates may have a holding in any of the securities discussed in this report; for securities where the holding is 1% or greater, the specific holding is disclosed in the Important Disclosures section above. **India:** For private circulation only, not for sale. **Pakistan:** For private circulation only, not for sale. **New Zealand:** This material is issued and distributed by JPMSAL in New Zealand only to persons whose principal business is the investment of money or who, in the course of and for the purposes of their business, habitually invest money. JPMSAL does not issue or distribute this material to members of "the public" as determined in accordance with section 3 of the Securities Act 1978. The recipient of this material must not distribute it to any third party or outside New Zealand without the prior written consent of JPMSAL. **Canada:** The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, or solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. The information contained herein is under no circumstances to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed judgment upon these materials, the information contained herein or the merits of the securities described herein, and any representation to the contrary is an offence. **Dubai:** This report has been issued to persons regarded as professional clients as defined under the DFSA rules.

General: Additional information is available upon request. Information has been obtained from sources believed to be reliable but JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) do not warrant its completeness or accuracy except with respect to any disclosures relative to JPMS and/or its affiliates and the analyst's involvement with the issuer that is the subject of the research. All pricing is as of the close of market for the securities discussed, unless otherwise stated. Opinions and estimates constitute our judgment as of the date of this material and are subject to change without notice. Past performance is not indicative of future results. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. The recipient of this report must make its own independent decisions regarding any securities or financial instruments mentioned herein. JPMS distributes in the U.S. research published by non-U.S. affiliates and accepts responsibility for its contents. Periodic updates may be provided on companies/industries based on company specific developments or announcements, market conditions or any other publicly available information. Clients should contact analysts and execute transactions through a J.P. Morgan subsidiary or affiliate in their home jurisdiction unless governing law permits otherwise.

"Other Disclosures" last revised September 29, 2012.

Copyright 2012 JPMorgan Chase & Co. All rights reserved. This report or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan.